

Appendix: Byzantine-Robust Federated Learning with Variance Reduction and Differential Privacy

A. Proof: Variance Decomposition

We measure the concentration property of the aggregated local updates in DPFed by observing its variance: According to the model training protocol of DPFed, we have

$$\Delta_i^t = \text{Clip}_C \left(\eta \sum_{r=0}^{\tau-1} \mathbf{g}_i^{t,r} \right) + \mathcal{N}(0, C^2 \sigma^2 \cdot \mathbf{I}_d). \quad (1)$$

Given that

$$\mathbb{E} \left[\frac{1}{s} \sum_{i \in \mathcal{S}^t} \Delta_i^t \right] = \frac{1}{n} \sum_{i \in [n]} \mathbb{E} \left[\text{Clip}_C \left(\eta \sum_{r=0}^{\tau-1} \mathbf{g}_i^{t,r} \right) \right] \quad (2)$$

The expectation $\mathbb{E}[\cdot]$ is taken over the sampled clients $\mathcal{S}^t$ and mini-batches $\xi_i^r, \forall i \in [n], r \in \{0, \dots, \tau-1\}$ at round t and the DP noise. Let $\zeta_i^t = \min(1, C/\|\eta \sum_{r=0}^{\tau-1} \mathbf{g}_i^{t,r}\|_2)$, one yield

$$\mathbb{E} \left[\frac{1}{s} \sum_{i \in \mathcal{S}^t} \Delta_i^t \right] = \frac{\eta}{n} \sum_{i \in [n]} \mathbb{E} \left[\zeta_i^t \sum_{r=0}^{\tau-1} \mathbf{g}_i^{t,r} \right] = \frac{C}{n} \sum_{i \in [n]} \mathbb{E} \left[\frac{\sum_{r=0}^{\tau-1} \mathbf{g}_i^{t,r}}{\|\sum_{r=0}^{\tau-1} \mathbf{g}_i^{t,r}\|_2} \right], \quad (3)$$

if we always use a small clipping threshold, i.e., $C < \|\eta \sum_{r=0}^{\tau-1} \mathbf{g}_i^{t,r}\|_2$.

For ease of expression, we consider the case of $\tau = 1$ and full participation so that the local gradient of each client will be directly sent to the server. Let $\mathbf{g}_i^t$ denote the local gradient of client i in round t , we have

$$\Delta_i^t = \text{Clip}_C (\mathbf{g}_i^t) + \mathcal{N}(0, C^2 \sigma^2 \cdot \mathbf{I}_d). \quad (4)$$

Let $\nabla f(\mathbf{x}^t)$ be the true gradient of $f(\mathbf{x}^t)$ and $\Delta^t = \frac{1}{n} \sum_{i \in [n]} \Delta_i^t$ as an estimator of $\nabla f(\mathbf{x}^t)$, the discrepancy between the estimator and the true gradient is decomposed as

$$\begin{aligned} \mathbb{E} \|\Delta^t - \nabla f(\mathbf{x}^t)\|^2 &= \mathbb{E} \left\| \frac{1}{n} \sum_{i \in [n]} \Delta_i^t - \frac{1}{n} \sum_{i \in [n]} \nabla f_i(\mathbf{x}^t) \right\|^2 \\ &= \mathbb{E} \left\| \frac{1}{n} \sum_{i \in [n]} \zeta_i^t \mathbf{g}_i^t - \frac{1}{n} \sum_{i \in [n]} \nabla f_i(\mathbf{x}^t) \right\|^2 + \frac{dC^2 \sigma^2}{s} \\ &= \mathbb{E} \left\| \frac{1}{n-B} \sum_{i \in \mathcal{H}} \zeta_i^t \mathbf{g}_i^t + \frac{1}{B} \sum_{i \in \mathcal{B}} \zeta_i^t \mathbf{g}_i^{t*} - \frac{1}{n} \sum_{i \in [n]} \nabla f_i(\mathbf{x}^t) \right\|^2 + \frac{dC^2 \sigma^2}{s} \\ &\leq 2\mathbb{E} \left\| \frac{1}{n-B} \sum_{i \in \mathcal{H}} \zeta_i^t \mathbf{g}_i^t - \frac{1}{n} \sum_{i \in [n]} \nabla f_i(\mathbf{x}^t) \right\|^2 + 2\mathbb{E} \left\| \frac{1}{B} \sum_{i \in \mathcal{B}} \zeta_i^t \mathbf{g}_i^{t*} \right\|^2 + \frac{dC^2 \sigma^2}{s} \\ &\leq 4\mathbb{E} \left\| \frac{1}{n-B} \sum_{i \in \mathcal{H}} (\zeta_i^t \mathbf{g}_i^t - \nabla f_i(\mathbf{x}^t)) \right\|^2 + 4\mathbb{E} \left\| \frac{1}{B} \sum_{i \in \mathcal{B}} \nabla f_i(\mathbf{x}^t) \right\|^2 + 2C^2 + \frac{dC^2 \sigma^2}{s} \\ &= \frac{4}{n-B} \sum_{i \in \mathcal{H}} \mathbb{E} \|\zeta_i^t \mathbf{g}_i^t - \nabla f_i(\mathbf{x}^t)\|^2 + 4\kappa^2 + 2C^2 + \frac{dC^2 \sigma^2}{s} \end{aligned}$$

Applying momentum, sparsification and clipping during model training processing can lead to a reduction in variances,

$$\begin{aligned} & \mathbb{E} \left\| \frac{1}{n} \sum_{i \in [n]} \text{Top}_k(\Delta_i^t) - \frac{1}{n} \sum_{i \in [n]} \nabla f_i(\mathbf{x}^t) \right\|^2 \\ & \leq 2\mathbb{E} \left\| \frac{1}{n-B} \sum_{i \in \mathcal{H}} \text{Top}_k(\zeta_i^t \mathbf{g}_i^t) - \frac{1}{n} \sum_{i \in [n]} \nabla f_i(\mathbf{x}^t) \right\|^2 + 2\mathbb{E} \left\| \frac{1}{B} \sum_{i \in \mathcal{B}} \text{Top}_k(\zeta_i^t \mathbf{g}_i^{t*}) \right\|^2 + \frac{kC^2\sigma^2}{s} \end{aligned}$$

Here, $\zeta_i^t = \min(1, C/\|\eta \mathbf{g}_i^{t,0}\|_2)$, and we assume the true local gradient is bounded, i.e., $\nabla f_i(\mathbf{x}) \leq \kappa^2$.

$$\Delta^t = \frac{1}{s} \sum_{i \in S^t} \eta \zeta_i^t \mathbf{g}_i^{t,0} + \mathcal{N}(0, (C^2\sigma^2/s) \cdot \mathbf{I}_d), \quad (5)$$

as the sum of Gaussian random variables is still a Gaussian random variable. Let $\nabla f(\mathbf{x}^t)$ be the true gradient of $f(\mathbf{x}^t)$ and Δ^t as an estimator of $\nabla f(\mathbf{x}^t)$, the discrepancy between the estimator and the true gradient is decomposed as

$$\begin{aligned} \mathbb{E} \|\Delta^t - \nabla f(\mathbf{x}^t)\|^2 &= \mathbb{E} \left\| \frac{1}{|S^t|} \sum_{i \in S^t} \Delta_i^t - \frac{1}{n} \sum_{i \in [n]} \nabla f_i(\mathbf{x}^t) \right\|^2 \\ &= \mathbb{E} \left\| \frac{\eta}{s} \sum_{i \in S^t} \zeta_i^t \mathbf{g}_i^{t,0} - \frac{1}{n} \sum_{i \in [n]} \nabla f_i(\mathbf{x}^t) \right\|^2 + \mathbb{E} \|\mathcal{N}(0, C^2\sigma^2 \mathbf{I}_d/s)\|^2 \\ &= \mathbb{E} \left\| \frac{\eta}{s} \sum_{i \in S^t} \zeta_i^t \mathbf{g}_i^{t,0} - \frac{1}{n} \sum_{i \in [n]} \nabla f_i(\mathbf{x}^t) \right\|^2 + \frac{dC^2\sigma^2}{s} \\ &= \mathbb{E} \left\| \frac{\eta}{|S_o^t|} \sum_{i \in S_o^t} \zeta_i^t \mathbf{g}_i^{t,0} + \frac{1}{|\mathcal{B}^t|} \sum_{i \in \mathcal{B}^t} b_i^t - \frac{1}{n} \sum_{i \in [n]} \nabla f_i(\mathbf{x}^t) \right\|^2 + \frac{dC^2\sigma^2}{s} \\ &\leq 2\mathbb{E} \left\| \frac{\eta}{|S_o^t|} \sum_{i \in S_o^t} \zeta_i^t \mathbf{g}_i^{t,0} - \frac{1}{n} \sum_{i \in [n]} \nabla f_i(\mathbf{x}^t) \right\|^2 + 2\mathbb{E} \left\| \frac{1}{|\mathcal{B}^t|} \sum_{i \in \mathcal{B}^t} b_i^t \right\|^2 + \frac{dC^2\sigma^2}{s} \\ &\leq 2\mathbb{E} \left\| \frac{\eta}{|S_o^t|} \sum_{i \in S_o^t} \zeta_i^t \mathbf{g}_i^{t,0} - \frac{1}{n} \sum_{i \in [n]} \nabla f_i(\mathbf{x}^t) \right\|^2 + \frac{2}{\beta n} \sum_{i \in \mathcal{B}} \mathbb{E} \|b_i^t\|^2 + \frac{dC^2\sigma^2}{s} \\ &\leq 2\mathbb{E} \left\| \frac{1}{|S_o^t|} \sum_{i \in S_o^t} \eta \zeta_i^t \mathbf{g}_i^{t,0} - \frac{1}{n} \sum_{i \in [n]} \eta \zeta_i^t \mathbf{g}_i^{t,0} + \frac{1}{n} \sum_{i \in [n]} \eta \zeta_i^t \mathbf{g}_i^{t,0} - \frac{1}{n} \sum_{i \in [n]} \nabla f_i(\mathbf{x}^t) \right\|^2 + \frac{2}{\beta n} \sum_{i \in \mathcal{B}} \mathbb{E} \|b_i^t\|^2 + \frac{dC^2\sigma^2}{s} \\ &\leq 4\eta^2 \mathbb{E} \left\| \frac{1}{|S_o^t|} \sum_{i \in S_o^t} \eta \zeta_i^t \mathbf{g}_i^{t,0} - \frac{1}{n} \sum_{i \in [n]} \zeta_i^t \mathbf{g}_i^{t,0} \right\|^2 + \frac{4}{n} \sum_{i \in [n]} \mathbb{E} \|\eta \zeta_i^t \mathbf{g}_i^{t,0} - \nabla f_i(\mathbf{x}^t)\|^2 + \frac{2}{\beta n} \sum_{i \in \mathcal{B}} \mathbb{E} \|b_i^t\|^2 + \frac{dC^2\sigma^2}{s} \\ &= 4\eta^2 \mathbb{E} \left\| \frac{1}{|S_o^t|} \sum_{i \in S_o^t} \eta \zeta_i^t \mathbf{g}_i^{t,0} - \frac{1}{n} \sum_{i \in [n]} \zeta_i^t \mathbf{g}_i^{t,0} \right\|^2 + \frac{4}{n} \sum_{i \in [n]} \mathbb{E} \|\eta \zeta_i^t \mathbf{g}_i^{t,0} - \nabla f_i(\mathbf{x}^t)\|^2 + \frac{2}{\beta n} \sum_{i \in \mathcal{B}} \mathbb{E} \|b_i^t\|^2 + \frac{dC^2\sigma^2}{s} \end{aligned}$$

where $\mathcal{B}^t$ represents the Byzantine clients selected in the t -th round, and $S_o^t = S^t - \mathcal{B}^t$ represents the selected benign clients.